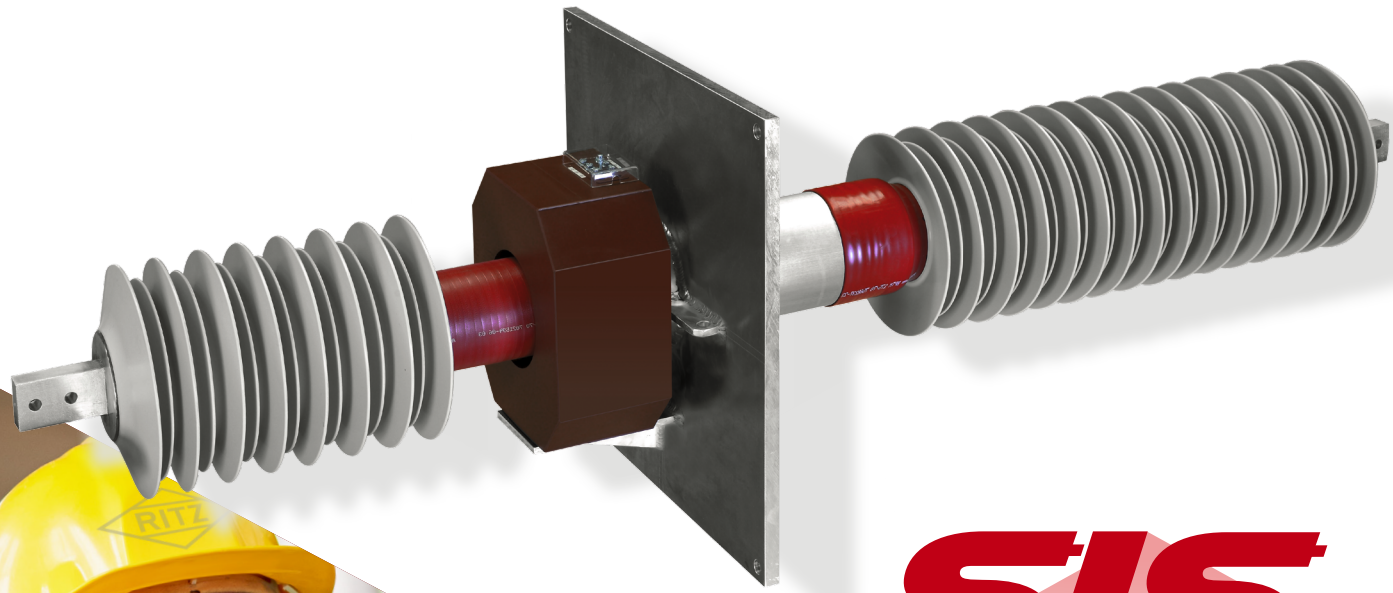
SISD BUSHINGS

FOR INDOOR AND OUTDOOR APPLICATIONS
UP TO 72.5 kV and 7000 A



SIS®

RITZ

SISD BUSHINGS

RITZ SIS PRODUCT RANGE

SIS BUS BAR SYSTEM
System voltage 1 kV up to 72.5 kV
System current 630 A up to 7000 A

SISD BUSHINGS
System voltage 3.6 kV up to 72.5 kV
System current 630 A up to 7000 A

DESIGN AND DESCRIPTION ELECTRICAL SAFETY

RITZ SISD bushings are insulated using Epoxy Resin Impregnated Paper (ERIP) technology with embedded half conductive control layers. The Bushings are designed and dimensioned according to the electro-technical requirements of the project.

Conductors are dimensioned according to the nominal current. They can be copper or aluminum and either solid or tubular. The conductors are then wrapped using electrical quality crepe paper. Semi-conductive capacitive grading layers are embedded during the wrapping process in order to control the electrical field. A grounding layer (earth) is embedded across the entire length. The grounding is connected to the fixing plate of the Bushing. Its function is to guarantee safe penetration of the wall or roof. The composite insulation is then impregnated with epoxy cast resin under vacuum in one of our autoclaves.

After manufacture, each Bushing is routine tested according to IEC60 137 in our high voltage test lab. This guarantees safe, partial discharge free operation. The design and subsequent high voltage testing of each bushing guarantees safety for human operators and longevity of our bushings.

TYPICAL FEATURES OF THE SISD BUSHINGS

- safe, available, reliable, compact
- very low maintenance
- easy installation
- long life span

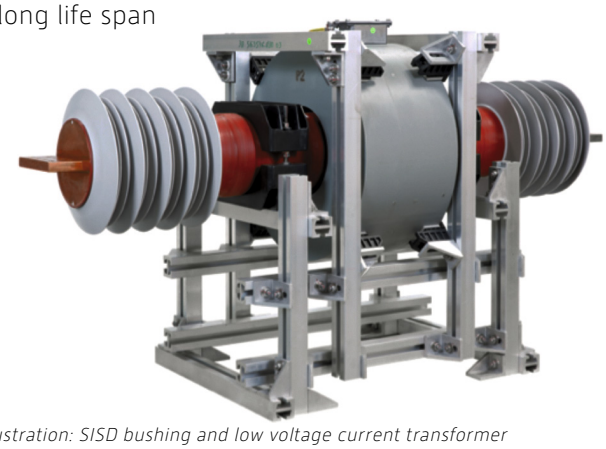
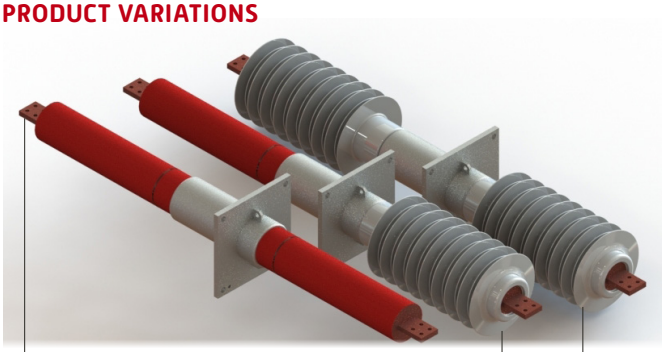


Illustration: SISD bushing and low voltage current transformer

PRODUCT VARIATIONS



MECHANICAL SAFETY

All bushings are protected both environmentally and mechanically by a high quality, compressed shrinking sleeve. The sleeve protects the insulation from humidity, dust and light mechanical impacts along the complete bushing length.The contact ring in the bushing terminals is designed with the same diameter as the insulation. In order to increase the creepage distance without adding additional bushing length we use silicone insulator sheds. We supply as standard 31 mm/kV SDC, according to IEC60815. This value is suitable for high polluted areas. Other levels are available on request.

It is possible to combine the SISD bushings with our ring core current tranformers as illustrated in the photograph below

TECHNICAL STANDARD DATA

International standard	IEC 60137:2017
Type	Capacitive graded bushing
Insulation	ERIP insulation (Epoxy Resin Impregnated Paper)
Conductor	E-AlMgSi0,5 or CuETP
Surface	Shrinking sleeve, compressed
Fixation plate	Aluminium, 20mm
Operating conditions	Indoor/Indoor, Indoor/Oudoor, Outdoor/Outdoor, located and fixed at wall, roof or ceiling
Altitute	<1000m *(others on request)*
Humidity	99%
Ambient temperature	-40 up to +40°C *(others on request)*
Mounting angle	0 to 90°
Insulation sheds	gray Silicone rubber
Creepage distance	Outdoor, min. 31mm/kV acc. IEC 60815 (others on request)
Wall Thickness	A) 300mm B) 500mm
Terminals	Flat or stud type
Max. conductor temp.	100°C at 40°C ambient
Mechanical impact resistance class	IK10

INDOOR

U_m / U_p / U_{BIL} [kV]	I_N from – up to [A]		Complete length A [mm]	Wall thickness A [mm]	Complete length B [mm]	Wall thickness B [mm]	Creepage distance [mm]	Sparking distance [mm]
3.6 /10 /40	630	6500	950	300	1150	500	177	175
7.2 /20 /60	630	6500	1010	300	1210	500	207	205
12 /28 /75	630	6500	1090	300	1290	500	247	245
17.5 /38 /95	630	6500	1140	300	1370	500	272	270
24 /50 /125	630	6500	1170	300	1370	500	287	285
36 /70 /170	630	6500	1330	300	1530	500	367	365
40.5 /85 / 200	630	6500	1330	300	1530	500	367	365
52 /95 /250	630	3150	1570	300	1770	500	487	485
72.5 /140 /325	630	3150	1880	300	2080	500	642	640

OUTDOOR

U_m / U_p / U_{BIL} [kV]	I_N from – up to [A]		Complete length A [mm]	Wall thickness A [mm]	Complete length B [mm]	Wall thickness B [mm]	Creepage distance [mm]	Sparking distance [mm]
3.6 /10 /40	630	6500	950	300	1150	500	587	241
7.2 /20 /60	630	6500	1010	300	1210	500	617	264
12 /28 /75	630	6500	1090	300	1290	500	657	297
17.5 /38 /95	630	6500	1140	300	1370	500	682	320
24 /50 /125	630	6500	1170	300	1370	500	807	334
36 /70 /170	630	6500	1330	300	1530	500	1147	410
40.5 /85 /200	630	6500	1330	300	1530	500	1297	407
52 /95 /250	630	3150	1570	300	1770	500	1677	530
72.5 /140 /325	630	3150	1880	300	2080	500	2462	685

I_N [A]	≤ 800		1000 – 1600		2000 – 2500		≥ 3150	
U_m [kV]	Cantilever operating load [N]	Cantilever test load [N]	Cantilever operating load [N]	Cantilever test load [N]	Cantilever operating load [N]	Cantilever test load [N]	Cantilever operating load [N]	Cantilever test load [N]
≤ 36	500	1000	625	1250	1000	2000	1575	3150
52	800	1600	800	1600	1000	2500	1575	3150
72.5	1000	2000	100	2000	1575	3150	2000	4000

Please contact for further information and quotations:

SIS SALES DEPARTMENT

Florian Daeumler
Phone: +49 9549 8915
florian.daeumler@ritz-international.com

Sascha Kensche
Phone: +49 9549 8993
sascha.kensche@ritz-international.com

Disclaimer: Errors and omissions excepted.
Subject to technical modifications. Images for reference only.

Empowering Our Electrical Future

GERMANY: RITZ HAMBURG | RITZ WIRGES | RITZ KIRCHAICH | RITZ DRESDEN
AUSTRIA: RITZ MARCHTRENK
HUNGARY: RITZ KECSKEMÉT
USA: RITZ HARTWELL | WAYNESBORO
CHINA: RITZ SHANGHAI

RITZ INSTRUMENT TRANSFORMERS GmbH

Location Kirchaich
Mühlberg 1
97514 Oberaurach-Kirchaich
Germany

Phone: +49 40 511 23 - 0
Fax: + 49 40 511 23 - 111
Email: info@ritz-international.com

We are the leading specialist for instrument transformers, cast resin parts and bus bars with cast resin insulation.

We not only develop your standard equipment, but also transform your ideas into customised products.
In exact accordance with your requirements!
For more information visit ***www.ritz-international.com*** or contact us at ***info@ritz-international.com***

www.ritz-international.com

07/2025